Project Design Phase
Proposed Solution

Date	3 February 2026
Team ID	LTVIP2026TMIDS79347
Project Name	Online payment fraud detection using machine learning
Maximum Marks	2 Marks

Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Online payment systems are highly vulnerable to fraudulent transactions, leading to financial losses and reduced customer trust. There is a need for a real-time intelligent system to accurately detect and prevent fraud.
2.	Idea / Solution description	Develop a machine learning–based fraud detection system that analyzes transaction amount, location, device, and user behavior to predict fraudulent transactions in real time.
3.	Novelty / Uniqueness	Unlike traditional rule-based systems, this solution uses adaptive machine learning models that continuously learn from new data and reduce false positives.
4.	Social Impact / Customer Satisfaction	Enhances trust in digital payments, reduces financial losses, and ensures a secure transaction experience for users and businesses.
5.	Business Model (Revenue Model)	Revenue can be generated by offering fraud detection as a subscription-based service to banks, fintech companies, and e-commerce platforms.
6.	Scalability of the Solution	The system can be deployed on cloud infrastructure and scaled to handle large transaction volumes using distributed computing and real-time processing.